

# Public Economics and Development

## Week 9: Targeting beneficiaries of public goods and services

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# Social Protection: Scale and Composition

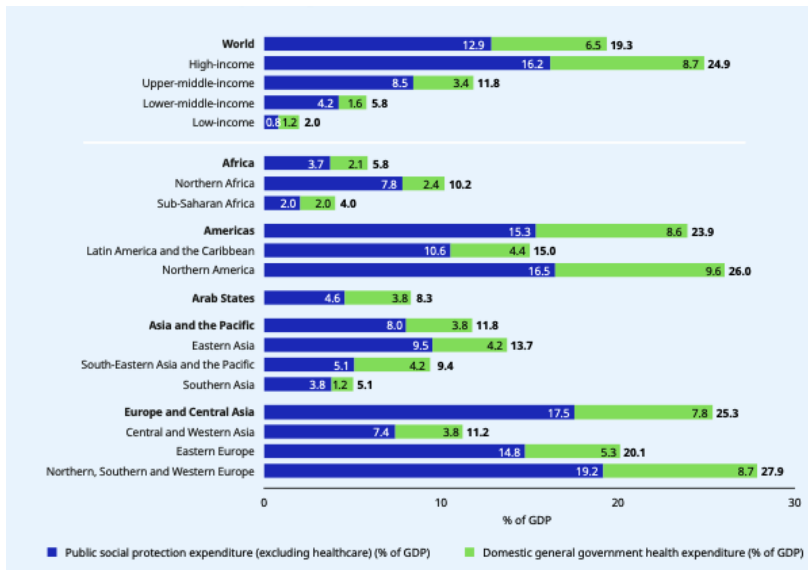
Growth of social protection programs (ILO World Social Protection Report 2024-26)

- **Old-age pensions:** 79.6% of people above retirement age receive a pension
- **Disability benefits:** 38.9% of people with severe disabilities receive a benefit
- **Child/family benefits:** 28.2% of children globally receive a family or child benefit
- **Unemployment benefits:** only 16.7% of unemployed people receive cash benefits
- **Others:** Insurance programs against climate change, Income support etc..

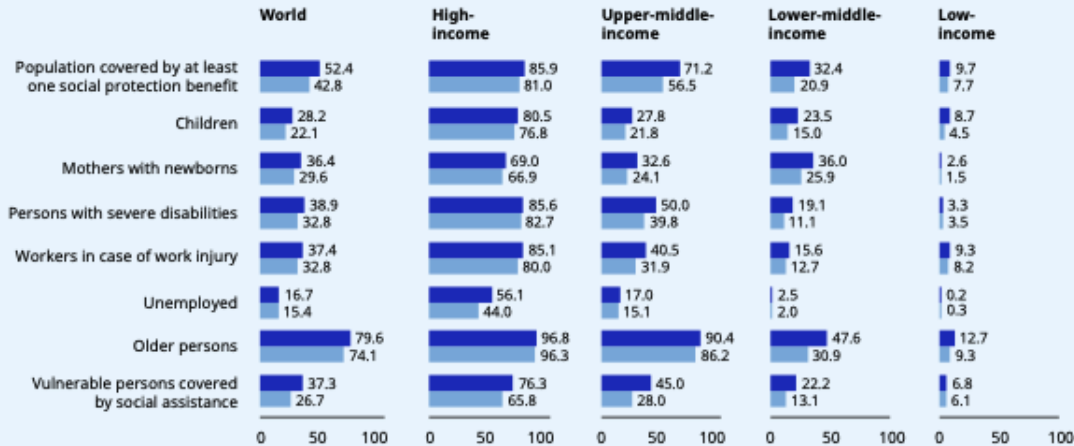
Overall, **52.4%** of the world's population are covered by at least one program

- But **3.8 billion people** remain uncovered
- Most of the uncovered are in low-income countries
  - ▶ Social protection spending averages only 3.7% of GDP vs. 12.9% globally.

# Variation in coverage under cash benefit programs



# Variation in population covered by social programs



# The Targeting vs. Universalism Debate

The case for targeting:

- Cost-effective: Concentrates resources on those who need them most
- Fiscal constraints make universal programs hard to sustain

The case for universalism:

- Identifying the poor can be administratively costly and error-prone
- Standard measures of poverty can be uninformative about actual welfare

Reality: Theoretical advantage of targeting can erode in practice

- Corruption and misallocation: Intended beneficiaries may not receive benefits
- Low-take up due to information gaps, stigma, or administrative barriers

# Why Don't Eligible Households Take Up Benefits?

Why take up is not complete among eligibles (Currie 2006)

- **Stigma:** “Shame” of receiving benefits,
- **Information barriers:** Those legally eligible unaware of their status or the procedure
- **Transaction costs:** Paperworks, or going to offices, can be costly and time-consuming
- **Social networks:** Fewer connections to prior recipients → less likely to participate

These costs may fall disproportionately on the neediest households

- Maybe even more time and cost constrained than typical households
- Could face higher barriers to the same information

⇒ Even a well-designed targeting rule may fail to reach its intended beneficiaries if these costs are ignored. *So how can we lower them?*

# Roadmap for today

## 1. Overview setup of the topic

## 2. Theories on targeting transfers

- 2.1 Nichols, Zeckhauser (1982): Targeting Transfers through Restrictions on Recipients
- 2.2 Kleven, Kopczuk (2011): Transfer Program Complexity and the Take-Up of Social Benefits

## 3. Empirical Evidence on targeting redistribution

- 3.1 Alatas, Banerjee, Hanna, Olken, Tobias (2012): Targeting the Poor: Evidence from a Field Experiment in Indonesia
- 3.2 Niehaus, Atanassova, Bertrand, Mullainathan (2013): Targeting with Agents
- 3.3 Castell, Gurgand, Imbert, Tochev (2025): Take-up of Social Benefits: Experimental Evidence from France

Nichols, Zeckhauser (1982):  
Targeting Transfers through Restrictions on  
Recipients



## Q: How to model social programs in presence of unobservable traits?

Government wants to redistribute to maximize social welfare

- Ideally, we base lump-sum tax based on random (and un-distortable) characteristics
- In practice, only the partial indicators of such traits are observed
  - ▶ Earnings: Is it because individuals are productive or they simply work long hours?
  - ▶ Expenditure: Reflection of true preference or other factors leading up to purchase?
- In such situations, we resort to second-best solutions using observable instruments

This paper provides a model on what instruments to use to target redistribution programs

- Target efficiency vs productive efficiency
- Introduces model for restrictions: Means-tested programs, in-kind transfers etc
- Compare its performance to programs solely relying on income tax and cash transfers

# Setting up the model: Income tax benchmark

## Setup of individual problems

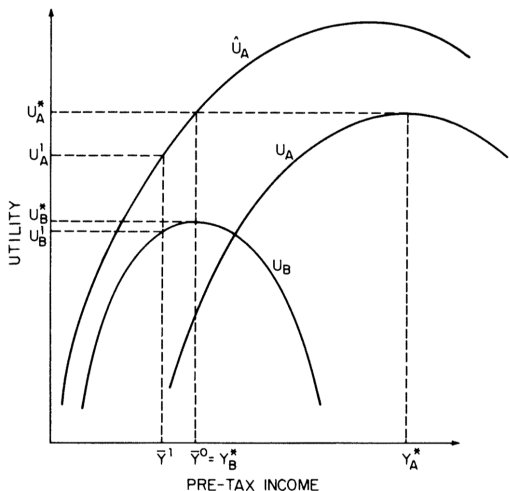
- We want high-wage individual  $A$  pay taxes to finance transfer to beneficiary  $B$
- Each consume  $C$  out of earned income  $Y$  and transfer  $T$ , thus  $C = Y + T$
- Each exert effort  $E$ , defined as earned income  $Y$  divided by wage  $W$
- Both individuals have same utility function  $U(C, E)$
- Government only observes income, and cannot break it down to wage and effort

## Solving the problem

- If  $Y > \bar{Y}$ , pay tax  $T$  (otherwise, receive transfer  $T$ )
- But  $A$  may choose to mimick  $B$ : Instead of  $Y_A^*$ , report  $\bar{Y}$  (to receive transfer)
- Hence the incentive compatibility constraint (Mimick by reducing effort, not ability)

$$U(Y_A^* - T, Y_A^*/W_A) > U(\bar{Y} + T, \bar{Y}/W_A)$$

# In income tax scheme, better to restrict eligibility



Two utilities for A

- $\hat{U}_A$  is utility of A mimicking B,  $U_A$  is utility of being truthful
- If  $\bar{Y} = Y_B^*$ , then A is indifferent between truthfully earning  $Y_A^*$  and mimicking  $\bar{Y}$

Slightly lower  $\bar{Y}$  to  $\bar{Y}^1$

- $U_B$  is reduced slightly: Productive efficiency  $\downarrow$
- $\hat{U}_A$  significantly falls at  $\bar{Y}^1 \rightarrow$  Better off at  $Y_A^*$
- Room to increase  $T$  without A mimicking and compensate loss in  $U_B$ : Targeting efficiency  $\uparrow$

FIGURE 1. DETERRENCE EFFECT OF INCOME CONSTRAINT

# In-kind Transfers

Is it optimal to give in-kind transfers instead of cash equivalent?

- In general, people (and economists) prefer cash transfers over in-kind
  - ▶ This is true when demand only depends on price and income
  - ▶ Exceptions: Commodity-specific externalities, paternalism etc
- IRL, demand may also be determined by leisure, ability, medical conditions etc
  - ▶ Even when income and price is held fixed, demand differs with the unobservable traits
  - ▶ Medical care consumption higher for those with poor health (even w/ income controlled)
- We refer to these goods as **indicator goods**

What are indicator goods in reality?

- Necessary staples (rice, cooking oils)
- Low-quality version of a good (e.g. housing)
- Health services

# In-kind transfers can be used as deterrence

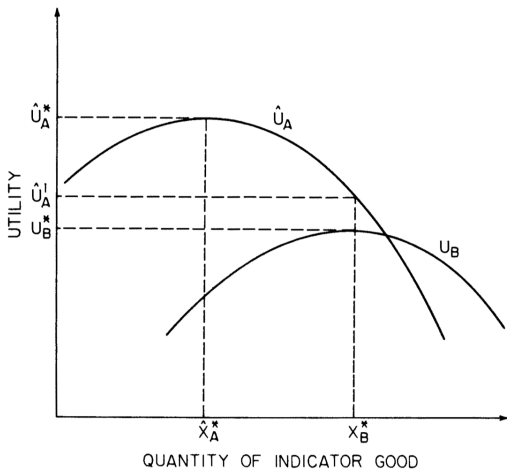


FIGURE 2. IN-KIND TRANSFERS TO DETER IMPOSTORS

Consumption of indicator good

- Out of B's income  $\bar{Y} + T$ , spend  $X_B^*$
- Mimicking A spend  $\hat{X}_A^*$  with the same income

Change the transfer scheme

- $\bar{X}$  is given in-kind, rest  $T - \bar{X}$  as cash
- If  $\bar{X} > \hat{X}_A^*$ ,  $\hat{U}_A \downarrow$ , and  $U_B \uparrow$  until  $\bar{X} = X_B^*$
- Better to make  $\bar{X} > X_B^*$ : B's marginal utility loss is small, but that of A is large
- This maximizes deterrence

# Ordeals and discussion

What are ordeals in this context?

- Deadweight loss that comes from eligibility restrictions
- Think qualification tests, administrative procedures, work requirements
- Effective costs from ordeals vary inversely with benefits
- This is because it works as a sorting mechanism

Discussion

- First-best solution is generally not possible (unobservable ability)
- Here, sacrificing the benefits for intended recipient may improve social welfare
- While it reduces welfare for the poor, it prevents unintended recipients
- However, how much of this is acceptable still remains an open debate

Kleven, Kopczuk (2011):  
Transfer Program Complexity and the Take-Up  
of Social Benefits

## Q: Optimal trade-off between complexity, eligibility, and generosity?

### Different dimensions of social programs

- Targeting (eligibility): Who is the intended recipient?
- Complexity: What hassles are involved in receiving the benefit?
  - ▶ Eligibility criteria, document requirements, trips to the office etc.
- Take-up: How many eligible beneficiaries do not receive the benefit?
  - ▶ Significant variation across different programs (Medicare vs disability assistance)

This paper provides a framework balancing these dimensions in designing welfare program

- Consider complexity as a program feature, rather than a side effect
- Complex programs w/ incomplete take-up can sometimes be desirable
- Study trade-off in false rejection (Type 1) and false awards (Type 2) error framework
  - ▶ Screen out undeserving recipients vs not deterring (and dropping) deserving applicants



# Setting up the model

## Definitions of classification errors

- Type 1a (1b) error: Eligible individuals do not apply (apply but wrongly rejected)
- Type 2 error: Ineligible individuals apply and accepted

## Other underlying parameters for individuals

- Each have ability  $a$ , with  $\sigma$  capturing precision by which this can be observed
- There is an indicator  $\tilde{a}$ , a signal of  $a$  defined as

$$\tilde{a} = a + \epsilon/\alpha \quad \epsilon \sim N(0, \sigma^2) \implies \epsilon/\sigma \sim N(0, 1)$$

where  $\epsilon$  is a noise  $\alpha$  represent complexity

- If eligibility criteria is  $\bar{a}$  and  $P(\cdot)$  represent distribution of  $\epsilon/\sigma$ , chance of receipt is

$$a + \epsilon/\alpha < \bar{a} \implies \epsilon/\sigma < \frac{\alpha(\bar{a} - a)}{\sigma} \implies P\left(\frac{\alpha(\bar{a} - a)}{\sigma}\right) = \bar{P}$$

# Modeling decision to apply

## Individual's decision to apply

- Utility depends on  $a$ , benefit  $B$ , and cost  $f(\alpha)$
- We can derive cutoff probability  $\tilde{P}$  that determines application

$$\bar{P}u(a + B - f(\alpha)) + (1 - \bar{P})u(a - f(\alpha)) > u(a) \iff \bar{P} > \frac{u(a) - u(a - f(\alpha))}{u(a + B - f(\alpha)) - u(a - f(\alpha))} \equiv \tilde{P}_a(\alpha, B)$$

- Individual applies when  $\bar{P} > \tilde{P}$
- Rise in  $\bar{a}$  always increases application, rise in  $a$  does the opposite
- Higher benefit  $B$  lowers  $\tilde{P}$  (easier to justify applying)
- When  $\bar{a} > a$ , rise in  $\alpha$  (complexity) and fall in  $\sigma$  (better precision) increase  $\bar{P}$
- When  $\bar{a} < a$  opposite conditions on  $\alpha$  and  $\sigma$  increase  $\bar{P}$

# Government problem

## Starting point

- At first, those with zero type 2 and type 1a error seems ideal
- But there is type 1b error: Imperfect testing  $\rightarrow$  eligible ones may be wrongly rejected
- We can reduce 1b error by increasing complexity, but increase 1a error
- Raising benefits can mitigate 1a error, but induces high fiscal burden

## Government goals

- Provide at least minimum income  $\bar{B}$  to as many eligible population on limited budget

$$\begin{aligned} & \max_{\alpha, \bar{a}, B} N_L(\alpha, \bar{a}, B) \\ & \text{s.t. } [N_L(\alpha, \bar{a}, B) + N_H(\alpha, \bar{a}, B)]B \leq R \text{ and } B \geq \bar{B} \end{aligned}$$

- $L$  types have low ability (otherwise  $H$  type)
- Not every  $L$  types (intended recipients) get support, and some  $H$  types get support

## Key result 1: If ability is identified precisely

Suppose we observe both  $a$  and  $\sigma$

- Full separation is possible in principle: No  $H$  types and only  $L$  types receive benefit
- If budget is small, provide  $\bar{B}$  to  $L$  groups with  $B = \bar{B}$ ,  $a_L < \bar{a} \leq a_H$
- If very large, set  $\bar{a}$  large and move towards universal provision and  $B \geq \bar{B}$

If budget is intermediate (first-best not achievable):

- Full separation requires  $B > \bar{B}$ ,  $\bar{a} = a_H$ , and high  $\alpha$  (Proposition 2)
  - ▶  $\alpha \uparrow$  to screen out  $H$  types +  $B \uparrow$  to compensate  $L$  types for the complexity (1a, 1b error  $\uparrow$ )
- But high  $\alpha \rightarrow L$  types deterred or wrongly rejected
- Forgoing full separation (let some  $H$  types in) strictly improves outcomes (Lemma 3)
  - ▶ Lower  $\alpha \rightarrow L$  types less discouraged +  $B$  no longer high to compensate for complexity
  - ▶ Savings from lower  $B$  outweigh the cost of the additional  $H$  type recipients
  - ▶ More  $L$  types can be served overall (at low benefits)

## Key result 2: Optimal programs feature all three errors

In intermediate budget, government has three tools to screen  $L$  from  $H$  types:  $\bar{a}$ ,  $\alpha$ ,  $B$

- **Set eligibility  $\bar{a}$  leniently:**  $H$  types allowed, but prevents discouragement of  $L$  types
  - ▶ Since full separation is no longer the goal,  $B$  can be set lower
  - ▶ Trades off: fewer Type 1a/1b errors, but more Type 2 errors
- **Use high complexity  $\alpha$**  to deter  $H$  types from applying
  - ▶ But this unavoidably discourages  $L$  types too
  - ▶ Trades off: fewer Type 2 errors, but more Type 1a and 1b errors
- **Raise benefits  $B$**  to persuade deterred  $L$  types back in to reduce type 1a error
  - ▶ Only worth doing when  $\bar{a}$  is set leniently - so that both  $H$  and  $L$  types are induced to join
  - ▶ Must be balanced against the budget cost of a higher  $B$

**Takeaway:** All three errors coexist in the optimum

( $\therefore$  each tool that reduces one type of error worsens another.)

## Key Takeaways

High complexity and incomplete take-up can be *optimal* policy outcomes

- Complexity is a screening tool, not merely a deadweight cost
- All three classification errors (Type 1a, 1b, Type 2) coexist in optimal programs
- Eliminating one type of error worsens another: trade-offs are unavoidable

Policy implication: Is it good to keep transaction costs high?

- Reducing transaction costs can increase take-up (we will see later)
- But the government may *rationaly* have chosen not to do so
- Screening benefits of complexity may outweigh the cost of incomplete take-up
- This justification weakens when factoring utility cost that complexity imposes
  - ▶ Likely discourage deserving applicants (who are also poor)
  - ▶ Note that no utility maximization is carried out here

Alatas, Banerjee, Hanna, Olken, Tobias (2012):  
Targeting the Poor: Evidence from a Field  
Experiment in Indonesia

## Q: How do different targeting mechanisms identify the beneficiaries?

Targeted social safety nets are increasingly becoming common

- In developed countries, this is usually done through means testing
- Many developing countries lack credible records to implement such mechanism
- Targeting that does not rely on observable income is in place
  - ▶ Proxy means test (PMT): Gathers info on assets and demographics that proxy income
  - ▶ Community-based targeting (CBT): Beneficiaries are selected by community members
- Prevent elite capture in communities vs better utilize local knowledge

This paper uses randomized experiments to examine which methods work better

- Compare PBT with different extent of CBT on 640 villages in Indonesia
- Measure performance on per capita expenditures and satisfaction with targeting
- Examine where the errors in judgement is from



# Institutional context

## Cash transfer program in Indonesia (Bantuan Langsung Tunai)

- Largest cash transfer in the developing world, started in '05 and renewed in '08
- Targeting done based on PMT and CBT methods
- Known to be targeted badly: 45% of nonpoor receives it, 47% of poor excluded
- Many were dissatisfied with the beneficiary lists (protests)

## Details of the experiment

- Provide IDR 30,000 (= \$3) to households below location-specific poverty line
- Government implemented the PMT in 1/3, CBT in 1/3, and hybrid in the rest
- In CBT villages, govt surveyed only selected households for means testing
- Overall number of beneficiaries set in advanced, revealed later

# Experimental design

## Main intervention

- T1: PMT villages - based on 49 indicators (housing attributes, assets, education etc)
- T2: CBT villages - Community meeting to rank hh by poverty, quota revealed later
- T3: Hybrid villages - Community ranking + verified by government officials
- Stratified randomization based on geography

## Subtreatments within CBT: Why do CBT differ from PMT?

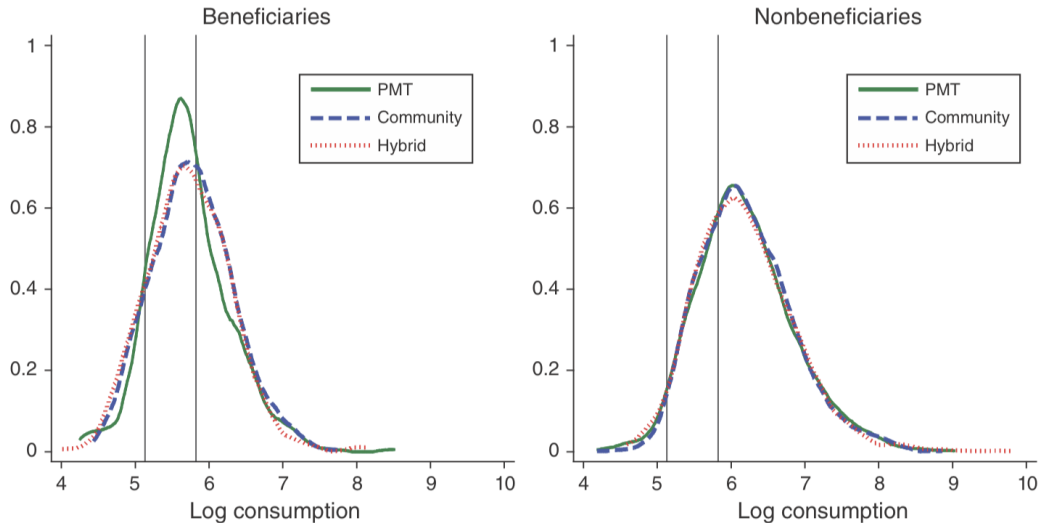
- Elite capture: Compare mechanism with all community members vs elites only
- Effort: Randomize order in which households are ranked to test if fatigue hurts efforts
- Preferences: Ranks determined by broadly shared preferences or particular ones
  - ▶ Vary meeting times (morning - women, evening - men, some meeting emphasize poverty)

# Errors included in CBT and hybrid treatments

TABLE 3—RESULTS OF DIFFERENT TARGETING METHODS ON ERROR RATE BASED ON CONSUMPTION

| Sample:               | Full population<br>(1) | By income status       |                        | By detailed income status |                      |                  |                   | Per capita consumption of beneficiaries<br>(8) |
|-----------------------|------------------------|------------------------|------------------------|---------------------------|----------------------|------------------|-------------------|------------------------------------------------|
|                       |                        | Inclusion error<br>(2) | Exclusion error<br>(3) | Rich<br>(4)               | Middle income<br>(5) | Near poor<br>(6) | Very poor<br>(7)  |                                                |
| Community treatment   | 0.031*<br>(0.017)      | 0.046**<br>(0.018)     | 0.022<br>(0.028)       | 0.028<br>(0.021)          | 0.067**<br>(0.027)   | 0.49<br>(0.038)  | −0.013<br>(0.039) | 9.933<br>(18.742)                              |
| Hybrid treatment      | 0.029*<br>(0.016)      | 0.037**<br>(0.017)     | 0.009<br>(0.027)       | 0.020<br>(0.020)          | 0.052**<br>(0.025)   | 0.031<br>(0.037) | −0.008<br>(0.037) | −1.155<br>(19.302)                             |
| Observations          | 5,753                  | 3,725                  | 2,028                  | 1,843                     | 1,882                | 1,074            | 954               | 1,719                                          |
| Mean in PMT treatment | 0.30                   | 0.18                   | 0.52                   | 0.13                      | 0.23                 | 0.55             | 0.48              | 366                                            |

# CBT does poor in capturing households just near the cutoff



# Elite capture does not(!) explain error rates

|                                                      | Attendance<br>(survey data) | Full sample<br>error rate | Full sample error rate |                   | On beneficiary list  |                      |
|------------------------------------------------------|-----------------------------|---------------------------|------------------------|-------------------|----------------------|----------------------|
|                                                      | (1)                         | (2)                       | (3)                    | (4)               | (5)                  | (6)                  |
| Community treatment                                  | 0.367***<br>(0.038)         | 0.029<br>(0.018)          | 0.033<br>(0.023)       | 0.048*<br>(0.025) | 0.042*<br>(0.025)    | 0.054*<br>(0.028)    |
| Hybrid treatment                                     | 0.370***<br>(0.037)         | 0.027<br>(0.018)          | 0.024<br>(0.022)       | 0.008<br>(0.024)  | 0.025<br>(0.022)     | 0.012<br>(0.023)     |
| Elite subtreatment                                   | -0.301***<br>(0.034)        | 0.004<br>(0.016)          | 0.016<br>(0.020)       | -0.013<br>(0.029) | -0.015<br>(0.021)    | -0.039<br>(0.032)    |
| Elite × hybrid                                       |                             |                           |                        | 0.062<br>(0.041)  |                      | 0.051<br>(0.043)     |
| Elite connectedness                                  |                             |                           | -0.025<br>(0.021)      | -0.025<br>(0.021) | -0.063***<br>(0.021) | -0.063***<br>(0.021) |
| Elite connectedness<br>× community treatment         |                             |                           | -0.015<br>(0.035)      | -0.013<br>(0.038) | -0.067**<br>(0.033)  | -0.078**<br>(0.036)  |
| Elite connectedness<br>× hybrid treatment            |                             |                           | 0.010<br>(0.033)       | 0.010<br>(0.035)  | -0.013<br>(0.033)    | -0.001<br>(0.035)    |
| Elite connectedness<br>× elite treatment             |                             |                           | -0.029<br>(0.031)      | -0.034<br>(0.047) | 0.041<br>(0.030)     | 0.064<br>(0.042)     |
| Elite connectedness<br>× elite treatment<br>× hybrid |                             |                           |                        | 0.003<br>(0.063)  |                      | -0.047<br>(0.060)    |
| Observations                                         | 287                         | 5,753                     | 5,753                  | 5,753             | 5,756                | 5,756                |
| Mean in PMT treatment                                | 0.11                        | 0.30                      | 0.30                   | 0.30              | 0.28                 | 0.28                 |

## Key takeaways and discussion

So what is happening with the CBT methods?

- Preference also does not play a role: no difference in outcomes due to meeting timing
- Communities apply standard of welfare different from per capita consumption
- Instead, villages use different metrics
  - ▶ Education, widowhood, presence of members not in the household, household size
  - ▶ Make guesses on households' ability to smooth consumption based on observables

Discussions

- Little reason to advocate hybrid: Worse of the two worlds
- PMT matches poverty based on consumption better
- CBT captures measures of ability to absorb shocks not reflected in consumption
- Satisfaction in targeting higher in CBT - result of reciprocity or other channels?

Niehaus, Atanassova, Bertrand, Mullainathan  
(2013):  
Targeting with Agents

## Q: How to achieve optimal targeting under corruptible agency?

Proxy means test (PMT) is used in places where reliable data on income is absent

- Ideally: Marginal reduction in poverty = Marginal admin cost
- However, many PMT departs from this rule in practice
- Leading explanation behind such deviation is corruption
  - ▶ Diverting transfers from intended recipients
  - ▶ Inflate claims about program participation
  - ▶ Demand bribes to issue permits to eligible or ineligible recipients

This paper provides a framework on targeting rule with possible corruption

- Principal define a targeting rule to be implemented by an agent
- Agents may have allocative preference, and possibly prefer bribes
- Principal cannot fully discipline these agents
- The paper provides properties of optimal targeting rules in this context
- Finds that more conditioning may *worsen* targeting



## Model Setup: Principal, Agent, and Households

Households are characterized by:

- Income  $y_i \in \{\underline{y}, \bar{y}\}$  (poor or rich) and observable characteristics  $x_i \in X$  (e.g. own TV?)
- Willingness to pay  $v_i$  for a program slot

Principal sets a targeting rule  $R \subseteq X$ :

- Household  $i$  is eligible iff  $x_i \in R$  ( $R$ : Eligible if no TV and no brick walls )
- Cannot observe income directly; must rely on proxy characteristics
- Has progressive preferences: values transfers to poor more than rich

Agent implements the rule and may deviate:

- Observes  $(y_i, x_i)$  and sets household-specific bribe prices  $p_i \geq 0$
- Has own allocative preferences over poor and rich households
- Rule violations detected with probability  $\pi$  and punished by fine  $f$
- Principal cannot make  $f$  arbitrarily large — enforcement is bounded

## More Targeting Can Backfire: The Key Intuition

Adding an eligibility indicator has two effects that pull in opposite directions:

- (+) **Statistical effect:** the rule better separates poor from rich households
- (-) **Enforcement effect:** verifying *ineligibility* becomes harder when it requires checking more criteria — officials grow less afraid of breaking the rule (i.e. become corrupt)

**Concrete example:** Suppose households with paved floors are ineligible.

- A third party can verify ineligibility by simply observing the floor  $\Rightarrow$  official is deterred
- Now refine the rule: ineligible only if paved floor *and* a television
- Statistically better, but verifying ineligibility now requires checking *two* things
- Official is less deterred  $\Rightarrow$  more rule violations among ineligible households

If the enforcement effect is strong enough, it can *more than offset* the statistical gain.

# When Conditions Fail: Simpler Rules Can Dominate

Typical setup in capable countries

- Strong enforcement: More tools available to verify eligibility
- Preference aligned: Principal and agent have same incentives
- Thus, gap in principal and agent objectives can be ignored

When none of these hold, the implications are stark:

- Fewer, easier-to-verify criteria  $>$  statistically optimal rule
- Sacrifices some statistical accuracy, but gains in enforceability can compensate
- At the extreme, universal eligibility can provide better welfare than accurate rule

**Key policy implication:** When implementation is corruptible, more information may hurt

- Trade-off between statistical vs. enforcement effect

# Empirical testing: India's BPL card

## History of public programs in India

- Originally universal public distribution system
- Switched to poverty targeting in 1997 citing fiscal costs
- Houses are now categorized into below the poverty line (BPL) or not
  - ▶ If BPL, given cards providing access to various public services

## Eligibility to BPL cards in Karnataka (research setting)

- Income, telephone, automobile, gas, color TV, own land, water pump etc
- State runs survey to check eligibility, implemented by village officials
- Researchers observe statutory eligibility + BPL receipt across 14,000 households
- Finds widespread rulebreaking: 48% of households misclassified
  - ▶ 70% of ineligible households have BPL
  - ▶ 13% of eligible households do not have BPL

# Actual distribution not progressive as statutory eligibility

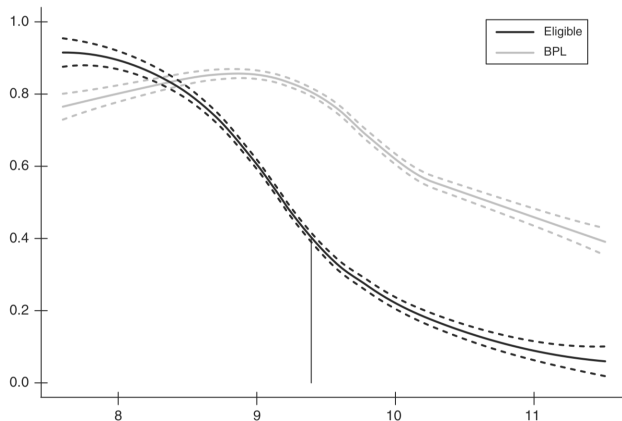


FIGURE 2. ELIGIBILITY IS MORE PROGRESSIVE THAN BPL STATUS

- Correlation with log income: -0.55 for statutory vs -0.23 for actual

# Key findings and discussion

## Key messages

- Using more indicators can backfire when implementation can be corruptible
- Rule simplicity has value: Simpler rules are harder to manipulate
- In such settings, trade-offs between accurately capturing poverty and enforcing rules

## Discussion

- The case for targeting over universalism is weaker than standard analysis suggests
- Particularly when enforcing complex rules are difficult
- Corruptible or incapable agency unravels benefits of accurate targeting

# Castell, Gurgand, Imbert, Tochev (2025): Take-up of Social Benefits: Experimental Evidence from France

## Q: How to increase take-up of welfare programs?

Many social programs suffer from low take-up

- 25% of eligible EITC beneficiaries do not receive them
- France: 36% of those eligible for out-of-work minimum income do not claim it
- Lack of information, transaction costs, welfare stigma often cited as reasons
- Also, does alleviating barriers attract poorer or richer claimants?

This paper uses 2 nationwide experiments to study barriers to take-up of welfare programs

- Focusing on job-seekers in France
- Experiment 1: Meet social workers to learn about eligibility and help with applications
- Experiment 2: Job-seekers learn about eligibility from online simulator
- Experiment 1 increased take-up far more than experiment 2
  - ▶ Suggesting that transaction cost is a bigger barrier than lack of information



# Experiment design

## Experiment 1: Meeting with social workers (RDVDE)

- Sample: 60,000 job-seekers with small or no claims to unemployment benefits
- Randomly select half of them to meeting with social services - 21% of them joined
- Discuss eligibility to 15 different benefits and support in applications
- Additional variation: Mode (In-person vs phone), Content (anti-stigma, info flyer)

## Experiment 2: Internet simulator on benefit eligibility

- Implemented one year before Experiment 1
- Sample: Randomly select among 40,000 job-seekers - 18% of treated group joined
- Provide personalized guidance on eligibility to same range of benefits
- Provides information, but do not lower transaction costs (no application help)

# Additional data and regression

## Additional data

- Job seekers' demographics from Pôle Emploi and CNAF
- Benefit receipt status from RNCPS and CNAF source
- Meeting outcomes and phone surveys conducted with French government

Regression leverages the randomized variation in treatment assignment status

- Intent-to treat regression (purely based on assignment)

$$Y_i = \beta T_i + \delta X_i + \pi_s + e_i$$

- Actual effect of attendance ( $M_i$ ): 2SLS regression

$$M_i = \gamma T_i + \alpha X_i + \pi_s + u_i$$

$$Y_i = \beta_{2SLS} \hat{M}_i + \delta X_i + \pi_s + \varepsilon_i$$

# Meetings increase takeup of benefits

|                                                                      | Any benefit        | Family benefit     | Health benefit     | Income benefit     |
|----------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|
| <i>Panel A. Effect of receiving an invitation to the RDVDE (ITT)</i> |                    |                    |                    |                    |
| Any invitation                                                       | 0.0174<br>(0.0037) | 0.0054<br>(0.0027) | 0.0017<br>(0.0023) | 0.0150<br>(0.0031) |
| Control group mean                                                   | 0.2669             | 0.1099             | 0.0761             | 0.1544             |
| Observations                                                         | 54,418             | 54,418             | 54,418             | 54,418             |
| <i>Panel B. Effect of attending the RDVDE (LATE)</i>                 |                    |                    |                    |                    |
| Attended meeting                                                     | 0.0815<br>(0.0172) | 0.0255<br>(0.0126) | 0.0079<br>(0.0106) | 0.0704<br>(0.0145) |
| Complier control mean                                                | 0.2859             | 0.1195             | 0.0897             | 0.1618             |
| Observations                                                         | 54,418             | 54,418             | 54,418             | 54,418             |

# Simulator has little effects

|                    | Any benefit received over eight months |                     |                    |                    |
|--------------------|----------------------------------------|---------------------|--------------------|--------------------|
|                    | Any benefit                            | Family              | Health             | Income             |
| Any invitation     | 0.0029<br>(0.0064)                     | −0.0041<br>(0.0059) | 0.0058<br>(0.0048) | 0.0039<br>(0.0045) |
| Control group mean | 0.4944                                 | 0.3898              | 0.1286             | 0.1635             |
| Observations       | 40,000                                 | 40,000              | 40,000             | 40,000             |

# Drop in claim size → Higher share of relatively richer beneficiaries

|                                                         | Any<br>benefit      | Family<br>benefit    | Income<br>benefit   |
|---------------------------------------------------------|---------------------|----------------------|---------------------|
| <i>Panel A. Benefits take-up</i>                        |                     |                      |                     |
| Any invitation                                          | 0.0155<br>(0.0048)  | 0.0012<br>(0.0031)   | 0.0154<br>(0.0043)  |
| Control group mean                                      | 0.2310              | 0.0797               | 0.1721              |
| Observations                                            | 31,444              | 31,444               | 31,444              |
| <i>Panel B. Amounts received conditional on take-up</i> |                     |                      |                     |
| Any invitation                                          | -5.7192<br>(3.9593) | -10.5496<br>(6.6148) | -1.9391<br>(4.0261) |
| Control group mean                                      | 158.4604            | 171.3916             | 133.3288            |
| Observations                                            | 7,556               | 2,526                | 5,713               |
| <i>Panel C. Benefits amounts received</i>               |                     |                      |                     |
| Any invitation                                          | 1.2589<br>(1.1963)  | -0.6397<br>(0.7109)  | 1.8737<br>(0.9113)  |
| Control group mean                                      | 36.6055             | 13.6613              | 22.9442             |
| Observations                                            | 31,444              | 31,444               | 31,444              |

## Other key findings and discussion

To whom were the intervention the most effective?

- Those who were least likely to attend meetings on their own
  - ▶ Estimated using marginal treatment effects (MTE) - not covered in 1st class
- Effects of meeting large for components with large application costs (items to fill)
  - ▶ But not too large - health benefits require going through a completely different branch
- Overall, results point to transaction cost being the biggest barrier to take-up

Discussion on targeting

- The intervention did not improve targeting (claim size smaller)
  - ▶ Marginal claimants induced by meetings are relatively richer (smaller claims)
- The very poor — most deterred by transaction costs — remained hard to reach
- Targeting is a supply-side problem from the welfare provider
- Whereas take-up is a demand-side problem (beneficiaries)

# Unanswered questions

Targeting deserving recipients and delivering services are tricky

- If information on income and preference is available, targeting income support is ideal
- Very rarely the case: Many misclassification errors exist
- For developing countries: Patchy information on income and monitoring agents
  - ▶ Use proxy means test or community based measures: imperfect information
  - ▶ Difficult to monitor agents who mishandle allocation
  - ▶ How to improve information and personnel capacities in this setup?
- Even if targeting is appropriate, take-up of welfare may be low
  - ▶ High transaction costs for the recipient
  - ▶ Others document lack of information and welfare stigma
  - ▶ How do you increase take-up of welfare among those eligible?